



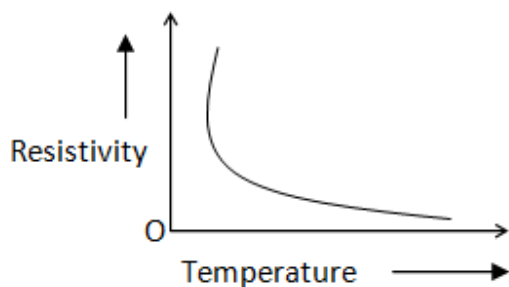
Q1. Define the term 'dielectric constant' of a medium in terms of capacitance of a capacitor.

Answer: We know that Capacitance of capacitor is defined as $C = \frac{\epsilon_0 \epsilon_r A}{d}$, $C_0 = \frac{\epsilon_0 A}{d}$, $C = \epsilon_r C_0$, $\epsilon_r = \frac{C}{C_0}$

Therefore **Dielectric constant** of a medium is defined as the ratio of the capacitance of the capacitor with the dielectric as the medium to its capacitance with vacuum between its plates.

Q2. Sketch a graph showing variation of resistivity of carbon with temperature.

Answer: The resistivity of carbon decreases with increasing temperature as shown:



Q3. The vertical component of Earth's magnetic field at a place is $\sqrt{3}$ times the horizontal component. What is the value of angle of dip at this place?

Answer: We know that angle of dip is defined as $\tan \delta = \frac{B_V}{B_H}$

Given: $B_V = \sqrt{3} B_H$

$$\tan \delta = \frac{\sqrt{3} B_H}{B_H} = \sqrt{3}$$

$\therefore$ Angle of dip, $\delta = 60^\circ$

Q4. With what purpose was famous Davisson-Germer experiment with electrons performed?

Answer: The purpose of Davisson-Germer experiment was to confirm the wave nature of electrons.

Q5. Name the type of communication in which the signal is a discrete and binary coded version of the message or information.

Answer: In Digital communication signal is binary in the form of 0 and 1.

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Q6. What are the laws of reflection?

Answer: Two Laws of reflections are

- (i) The angle of incidence (i) is equal to the angle of reflection (r) ($\angle i = \angle r$)
- (ii) The incident ray, the reflected ray and the normal lie in the same plane.

Q7. Name the physical quantity, whose SI unit is newton coulomb⁻¹.

Answer: It is the unit of electric field intensity ($E = F/q$).

Q8. State two factors by which the range of transmission of signals by a T.V. tower can be increased.

Answer: (I) By increasing the height of the tower.

(II) By increasing the height of the receiving antenna, so that it may directly intercept the signal from the transmitting antenna.

Q9. The electric field and electric potential at any point due to a point charge kept in air is 20 NC^{-1} and 10 JC^{-1} respectively. Compute the magnitude of this charge.

Answer: Given $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = 20 \text{ NC}^{-1}$ and $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = 10 \text{ JC}^{-1}$ therefore $E=V/r$
 $\therefore r = \frac{V}{E} = \frac{10}{20} = 0.5 \text{ m}$

Also Charge, $q = 4\pi\epsilon_0 rV = \frac{0.5 \times 10}{9 \times 10^9} = 0.55 \times 10^{-9} \text{ C}$ [$\because C = 4\pi\epsilon_0 r$]

Q10. Write the mathematical relation between mobility and drift velocity of charge carriers in a conductor. Name the mobile charge carriers responsible for conduction of electric current in (I) an electrolyte (II) an ionized gas.

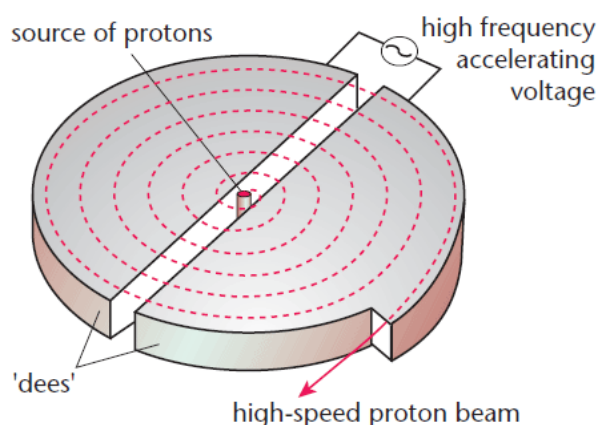
Answer: Mobility = $\frac{\text{Drift velocity}}{\text{Electric field}}$ or, $\mu = \frac{v_d}{E}$

- I. The charge carriers in an electrolyte are positive and negative ions.
- II. The charge carriers in an ionized gas are electrons and positively charged ions.



Q11. State the principle of working of a cyclotron. Write two uses of this machine.

Answer: **Principle:** Cyclotrons accelerate charged particle beams using a high frequency alternating voltage which is applied between two "D"-shaped electrodes. An additional static magnetic field B is applied in perpendicular direction to the electrode plane. Thus charge particle acquires very high kinetic energy with the help of relative small electric field.



Working: A cyclotron consists of two D-shaped regions known as dees. In each dee there is a magnetic field perpendicular to the plane of the page. In the gap separating the dees there is a uniform electric field pointing from one dee to the other. When a charge is released from rest in the gap it is accelerated by the electric field and carried into one of the dees. The magnetic field in the Dee causes the charge to follow a half-circle that carries it back to the gap.

While the charge is in the dee the electric field in the gap is reversed, so the charge is once again accelerated across the gap. The cycle continues with the magnetic field in the dees continually bringing the charge back to the gap. Every time the charge crosses the gap it picks up speed. This causes the half-circles in the dees to increase in radius, and eventually the charge emerges from the cyclotron at high speed.

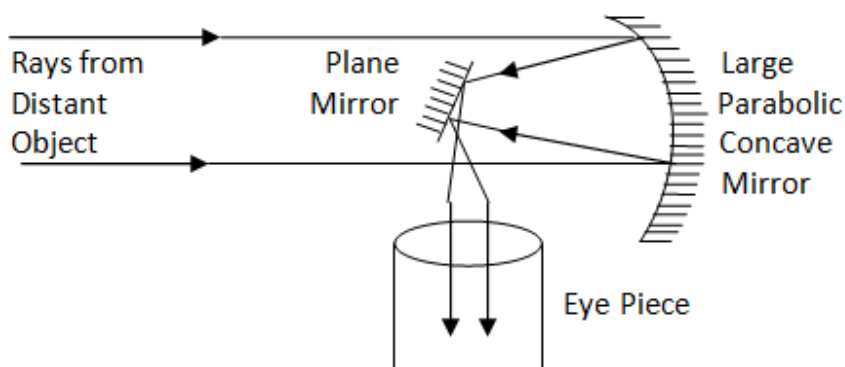
Uses: (I) Cyclotrons is used in particle therapy to treat cancer.

(II) A cyclotron is used to bombard nuclei with energetic particles and study the resulting nuclear reactions.



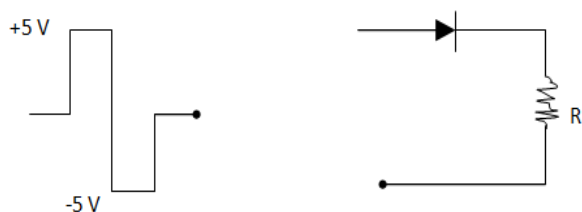
Q12. Draw a labeled ray diagram of a reflective type telescope. Write its one advantage over refracting type telescope.

Answer: The ray diagram of a reflecting type telescope:



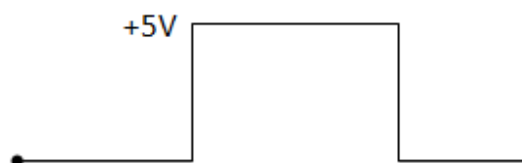
Advantage: Objective of reflecting telescope is a mirror (not lens), so the image formed is free from chromatic aberration (which occurs in lens as lens defect because of which image splits/spreads).

Q13. Draw and explain the output waveform across the load resistor R, if the input waveform is as shown in the given figure –



Answer: We know that diode gives output only when Anode (Collector) is at positive voltage and Plate (emitter) is at negative voltage, this we call as forward biased diode. In this given waveform when the input voltage is +5V, the diode gets forward biased, hence output across R is +5V will be obtained however during the next cycle of input waveform it clearly shows input voltage is -5V, the diode gets reversed biased so no output will be obtained across R.

Output across resistance R is as shown



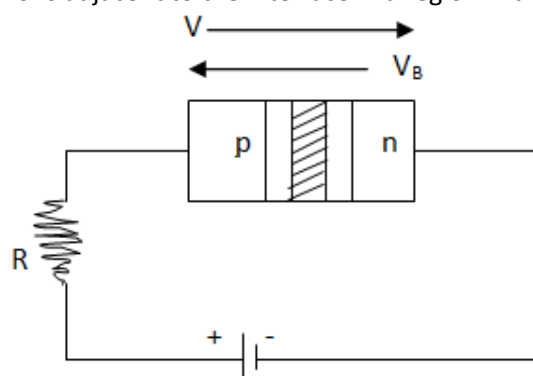


Or,

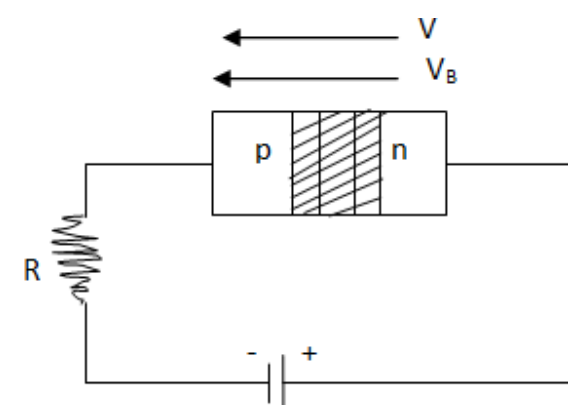
Explain how the width of depletion layer in a p-n junction diode changes when the junction is (I) forward biased (II) reverse biased.

Answer:

We know that N-type semiconductor has excess of free electrons and P-type has excess of holes. The diffused electrons come into contact with holes on the P-side and the diffused holes on the N-side. The net result is the diffused electrons and holes are gets neutralized due to reunion, leaving behind the charged ions adjacent to the interface in a region with no mobile carriers this is called the depletion region.



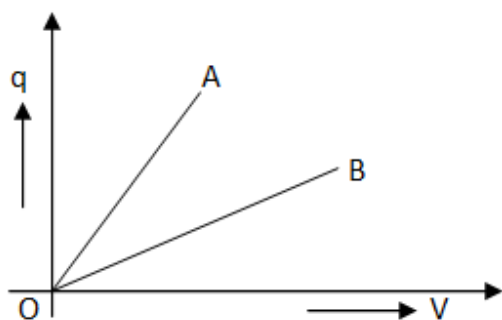
When a p-n junction is forward biased (p with +ve , n with –ve terminal of battery), the majority charge carries flow towards the junction, the width of the depletion layer decreases, as shown in the figure



When a p-n junction is reversed biased, p-type connected to –ve terminal and n-type connected to +ve terminal of battery, the majority charge carries flow away from the junction, the width of the depletion layer increases, as shown in the figure



Q14. The given graph shows the variation of charge q versus potential difference V for two capacitors C_1 and C_2 . The two capacitors have same plate separation but the plate area of C_2 is double than that of C_1 . Which of the lines in the graph correspond to C_1 and C_2 and why?

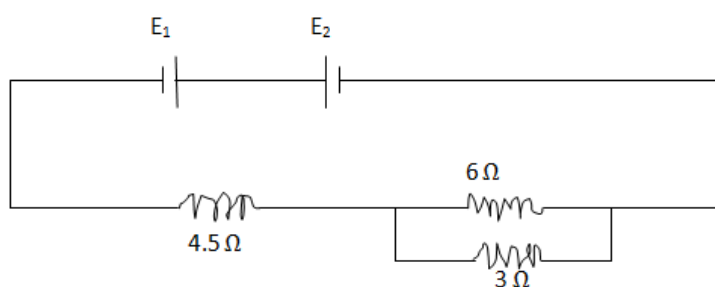


Answer: We know that $q = CV$ comparing with $y=mx$, here $m=C=\tan\theta$, higher is the slope higher is the capacitance. and graph A has a larger slope than B, so the graph A shows higher capacitance.

Also, $C = \frac{\epsilon_0 A}{d}$ i.e., $C \propto A$

As the plate area of C_2 is double of that of C_1 , so C_2 has a higher capacitance. Hence the line A of the graph corresponds to capacitor C_2 .

Q15. Two cells E_1 and E_2 in the given circuit diagram have an emf of 5 V and 9 V and internal resistance of 0.3Ω and 1.2Ω respectively. Calculate the value of current flowing through the resistance of 3Ω .



Answer: Net emf = $E_2 - E_1 = 9 - 5 = 4V$

Total resistance = $0.3 + 1.2 + 4.5 + \frac{6 \times 3}{6+3} = 8 \Omega$

Current through the circuit, $I = \frac{4}{8} = 0.5 A$, Let i_1 and i_2 be the current in 6 and 3Ω resistance

respectively. Therefore $i_1 + i_2 = 0.5$, as current remains same 0.5 in series with 4.5Ω resistance. Now Potential difference in parallel resistance is same hence $6i_1 = 3i_2$ or $i_1 = 0.5i_2$, so $0.5i_2 + i_2 = 0.5$ or $i_2 = 0.5/1.5 = 1/3 A$



Q16. How is the mutual inductance of a pair of coils affected when:

- (I) Separation between the coils is increased?
 - (II) The number of turns of each coil is increased?
 - (III) A thin iron sheet is placed between the two coils, other factors remaining the same?
- Explain your answer in each case.

Answer:

- (I) When the separation between the two coils is increased, flux that is lines of force cutting the secondary decreases so the mutual inductance decreases.
- (II) Mutual inductance is proportional to the product of number of turns in coils ($M \propto N_1 N_2$)
Hence Mutual inductance increases when the number of turns in each coil is increased,
- (III) Mutual inductance is proportional to the permeability $M \propto \text{permeability } (\mu)$. Iron sheet being ferromagnetic has higher permeability allows more lines of force (flux) to pass through , hence when an iron sheet is placed between the two coils the mutual inductance M increases.



Q17. A convex lens made up of glass of refractive index 1.5 is dipped, in turn, in:

- (I) Medium A of refractive index 1.65
- (II) Medium B of refractive index 1.33

Explain, giving reasons, whether it will behave as a converging lens or a diverging lens in each of these two media.

Answer:

- (I) Given refractive index of glass = 1.5, refractive index of medium A = 1.65. Therefore focal length f_A in medium A

$$f_A = \left(\frac{1.5}{1.65} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = - \frac{0.15}{1.65} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Since focal length f_A is negative hence the lens is diverging in nature.

- (II) Given refractive index of glass = 1.5, refractive index of the medium B = 1.33, therefore focal

$$\text{length } f_B \text{ in medium B } f_B = \left(\frac{1.5}{1.33} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{0.17}{1.33} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Since f_B is positive hence the lens is converging in nature.

Q18. Define the terms threshold frequency and stopping potential in relation to the phenomenon of photoelectric effect. How is the photoelectric current affected on increasing the (I) frequency (II) intensity of the incident radiations and why?

Answer:

Threshold frequency: The minimum frequency of incident radiation which can emit electrons from a metal is called *threshold frequency*. If the frequency of incident light is below the threshold frequency, photoelectric emission will not happen.

Stopping Potential: On heating metal plate by filament it emits photo electrons, to collect these photo electrons other plate (anode) kept at positive potential, however if we make collector at negative potential it will repel emitted electrons. The minimum negative potential given to the anode of a photocell for which the photoelectric current becomes zero is called *stopping potential*.

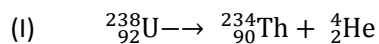
- (I) The increase of frequency of incident radiation has no effect on the photoelectric current.
- (II) The photoelectric current increases proportionally with the increase in intensity of incident radiation.



Q19. Explain, with the help of a nuclear reaction in each of the following cases, how the neutron to proton ratio changes during (I) alpha-decay (II) beta-decay.

Answer: We know that in alpha decay, mass number reduces by 4 units; atomic number reduces by 2 units. In beta decay atomic number increased by 1 unit but there is no change in mass number.

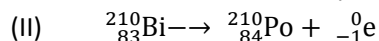
Number of Neutrons = Mass number – Number of proton (atomic number)



Neutron to proton ratio before α – decay = $\frac{238-92}{92} = \frac{146}{92} = 1.58$

Neutron to proton ratio after α – decay = $\frac{234-90}{90} = \frac{144}{90} = 1.6$

Thus the neutron to proton ratio increases in an α – decay.



Neutron to proton ratio before β – decay = $\frac{210-83}{83} = \frac{127}{83} = 1.53$

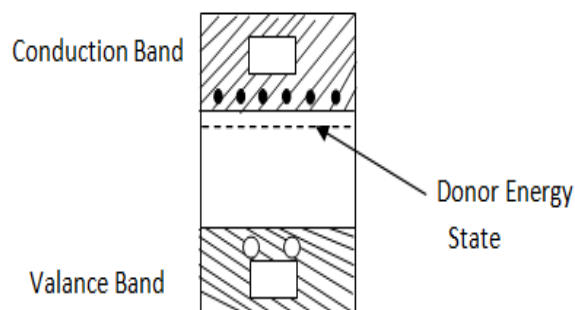
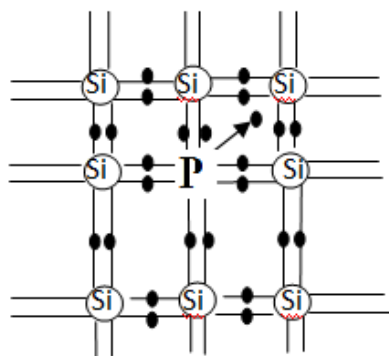
Neutron to proton ratio after β – decay = $\frac{210-84}{84} = \frac{126}{84} = 1.5$

Thus the neutron to proton ratio decreases in a β – decay.

Q20. What is an intrinsic semiconductor? How can this material be converted into (I) N-type (II) P-type extrinsic semiconductor? Explain with the help of energy band diagrams.

Answer: The naturally occurring pure semiconductor (without doping) where conductivity depends only on electrons thermally excited from valence band to conduction band is called an intrinsic semiconductor.

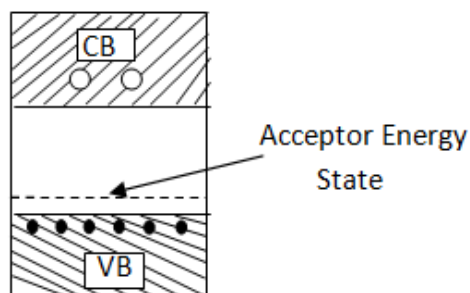
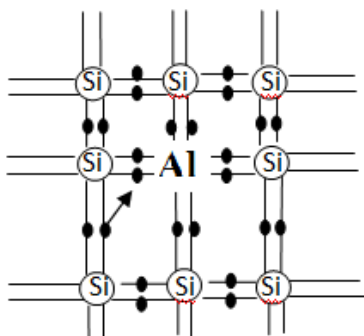
N-type semiconductor :In N-type semiconductor charge carrier is negative charge type i.e electron, when we dope silicon(Si), which has four valence electrons, with a controlled amount of penta-valent atoms like phosphorus atoms (P), which has five valence electrons, the atoms of the impurity element P will substitute the Si atoms.





Out of five valence electrons of Phosphorus (P) four electrons form covalent bonding with Silicon (Si) atom, while the fifth electron of Phosphorus is free to move. The penta-valent atoms are called the donor atoms because they donate electrons to the host crystal also since the free charge carrier is electron hence this *Extrinsic semi-conductor* is called n-type. On giving up their fifth electron, the donor atoms become positively charged. However, the material remains electrically neutral as a whole.

P-type semi-conductor: When Silicon atom Si (intrinsic semiconductor) is doped with in controlled amount by trivalent atoms like Aluminium (Al), Indium (In) or boron (B) atom, 3 valence electron of trivalent atom say Aluminium (Al) forms covalent bond with neighboring Silicon atom however due to deficiency of electron 1 incomplete covalent bond with Silicon atom remains.



This is completed by taking an electron from one of the Si-Si bonds, thus completing the Al-Si bond. This makes it ionized (negatively charged), and creates a hole or an electron deficiency in the covalently bonded electron system in the crystal as shown in the figure. The trivalent atoms are called *acceptor atoms* and this *extrinsic semi-conductor* is known as *p-type semi-conductor* as charge carrier is hole.

Q21. Why is the mass of a nucleus always less than the sum of the masses of its constituents, neutrons and protons? If the total number of neutrons and protons in a nuclear reaction is conserved, how then is the energy absorbed or evolved in the reaction? Explain.

Answer: Energy is required to hold positively charged proton together in nucleus. In a nucleus, the protons and neutrons come closer to a distance of 10^{-14} m. The energy required for the purpose is spent by the nucleons at the expense of their masses. The difference is mass converted to energy, so mass of the nucleus formed is less than the sum of the masses of the individual nucleons.

Here proton number and neutron number are conserved in a nuclear reaction; the total rest mass of neutrons and protons is the same on either side of a reaction. However total binding energy of nuclei on the left side need not be the same as that on the right hand side. The difference in these binding energies appears as energy released or absorbed in a nuclear reaction.

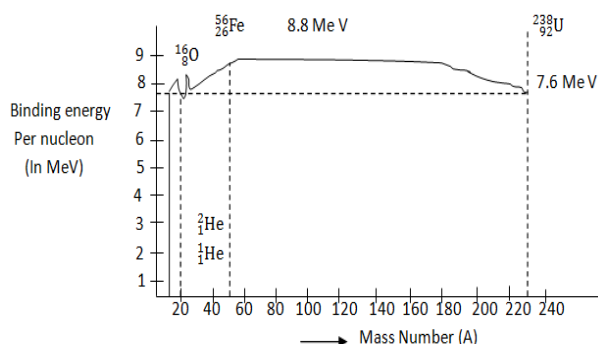


Or,

Draw a graph showing the variation of binding energy per nucleon with mass number for different nuclei.

Explain, with the help of this graph, the release of energy by the process of nuclear fusion.

Answer: **Binding Energy Curve:** The variation of average binding energy (B.E) per nucleon with mass number A is shown in the fig.



The above graphical presentation suggests:

- (I) Average B.E. per nucleon for light nuclei like ${}^1_1\text{H}$ and ${}^2_1\text{H}$ is small (2 to 3 MeV).
- (II) For mass number 2 to 20, the peaks are sharply defined. So that ${}^4_2\text{He}$, ${}^{12}_6\text{C}$, ${}^{16}_8\text{O}$, ${}^{20}_{10}\text{Ne}$ etc have more stability than the nuclei in their neighborhood.
- (III) From the graph most stable nucleus (e.g. ${}^{56}_{26}\text{Fe}$) having intermediate mass number 56 has **maximum** Binding energy as 8.8 Me V/N.
- (IV) The B.E. /nucleons decreases gradually after mass number ${}^{56}_{26}\text{Fe}$ to ${}^{238}_{92}\text{U}$. The heavy nuclei are relatively less stable.
- (V) Binding energy per nucleons is small for both light and heavy nuclei.
When **light nuclei fuse (nuclear fusion)** to form a heavy nucleus, high value of binding energy is released. Similarly when a **heavy nucleus splits (nuclear fission)** into lighter nuclei, **high** value of binding energy is released.

Q22. Define the term modulation. Name three different types of modulation used for a message signal using a sinusoidal continuous carrier wave. Explain the meaning of any of these.

Answer:

Modulation: It is the process in which some characteristic such as amplitude, frequency or phase angle of a high frequency carrier wave is changed in accordance with the instantaneous value of the low frequency modulating signal.

A sinusoidal carrier wave can be modulated in three ways:



- (I) Amplitude modulation
- (II) Frequency modulation
- (III) Phase modulation

Amplitude Modulation (AM): In AM process the modulating frequency wave is superimposed on high frequency carrier wave such that the frequency of the modulated wave remains the same as that of the carrier wave but its **amplitude** is modified in accordance with that of modulating frequency wave.

Frequency Modulation (FM): In FM process the modulating frequency wave is superimposed on high frequency carrier wave such that the amplitude of the modulated wave remains the same as that of the carrier wave but its **frequency** is modified in accordance with that of the audio frequency wave.

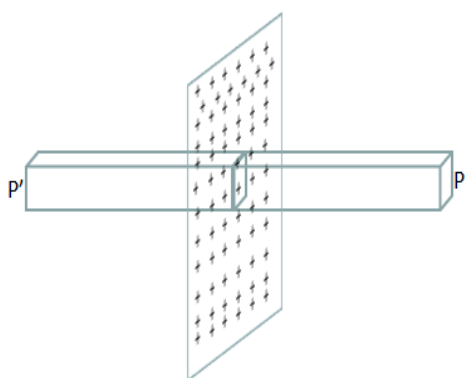
Q23. What is electric flux? Write its S.I. units. Using Gauss's theorem, deduce an expression for the electric field at a point due to a uniformly charged infinite plane sheet.

Answer: The electric flux through a given surface area is the total number of electric lines of force passing normally through this area. It is defined as the dot product of intensity and area, scalar quantity.

$$\Delta\phi = \vec{E} \cdot \vec{\Delta S}$$

The SI unit of electric flux = $\text{Vm}^{-1} \times \text{m}^2 = \text{Vm}$. [Since $E = V/d$, intensity is potential gradient unit of $E = V/m$]

Intensity at a point near an infinitely charged insulating plate:



Given : σ = surface density of charge i.e. charge per unit surface area

ϵ = permittivity of the surrounding medium

P is the given point close to the surface of the charged plate.

Let E be the intensity at P ?

Let us imagine a point P' opposite to P and at the same distance at P.

Let us imagine a rectangular box passing through the points P & P'. This closed surface is our Gaussian surface and is known as pill box.

Since the lines of force are emitted normally from the surface they are parallel to the two vertical and two horizontal faces of the Gaussian surface and hence flux through this faces will be zero. The lines of force are perpendicular to the two end faces only.



Hence the total flux through the Gaussian surface

$$\phi = 0 + 0 + 0 + 0 + EA + EA = 2EA \rightarrow (1)$$

Applying Gauss's theorem the flux through the Gaussian surface

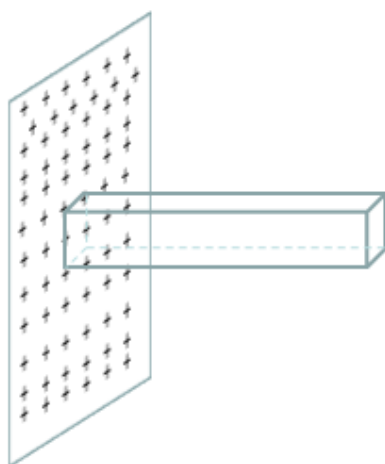
$$\phi = \frac{\text{charged enclosed}}{\text{permittivity}} = \frac{\sigma A}{\epsilon} \rightarrow (2)$$

From equation (1) and (2)

$$2EA = \frac{\sigma A}{\epsilon}$$

$$E = \frac{\sigma}{2\epsilon} \rightarrow (3)$$

Intensity at a point near an infinitely long charged conducting plate



Since lines of force can not exist inside a conductor we get lines of force only in the side of the charged face.

$$\phi = 0 + 0 + 0 + 0 + EA = EA \rightarrow (1)$$

$$\phi = \frac{\sigma A}{\epsilon} \rightarrow (2)$$

$$EA = \frac{\sigma A}{\epsilon}$$

$$E = \frac{\sigma}{\epsilon}$$



Q24. A 10 m long wire of uniform cross-section and $20\ \Omega$ resistance is used in a potentiometer. The wire is connected in series with a battery of 5 V along with an external resistance of $480\ \Omega$. If an unknown emf E is balanced at 6.0 m length of the wire, calculate:

- (I) The potential gradient of the potentiometer wire
- (II) The value of unknown emf E

Answer:

Given $L=10\text{ m}$, $R_{AB}=20\ \Omega$, $V = 5$, $l = 6\text{ m}$, $R = 480\ \Omega$

Let I = Current through the potentiometer wire AB then

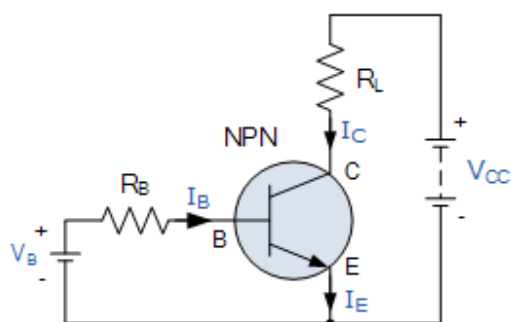
$$I = \frac{V}{R_{AB} + R} = \frac{5}{20 + 480} = 0.01\text{ A}$$

Potential difference across the wire AB $= V = IR_{AB} = 0.01 \times 20 = 0.2\text{ V}$

- (I) Potential gradient is $= \frac{V}{L} = \frac{0.2}{10} = 0.02\text{ Vm}^{-1}$
- (II) Since unknown emf balanced at 6 m length, therefore Potential drop across 6 m wire is equal to emf of the cell. Potential drop per unit length i.e. potential gradient $= 0.02$ so Potential drop across 6 m wire $= E = 0.02 \times 6 = 0.12\text{ V}$

Q25. The input resistance of a transistor is $1000\ \Omega$. On changing its base current by μA , the collector current increases by 2 mA. If a load resistance of $5\text{ K}\Omega$ is used in the circuit, calculate:

- (I) the current gain
- (II) Voltage gain of the amplifier.



Answer: Given $R_{in} = 1000\ \Omega$

$$\Delta I_B = 10\ \mu\text{A} = 10^{-5}\text{ A}$$

$$\Delta I_C = 2\text{ mA} = 2 \times 10^{-3}\text{ A}$$

$$R_L = 5\text{ k}\Omega = 5 \times 10^3\ \Omega$$

$$\text{Current gain, } \beta = \frac{\Delta I_C}{\Delta I_B} = \frac{2 \times 10^{-3}}{10^{-5}} = 200$$

$$\text{Voltage gain, } A_V = \beta \frac{R_L}{R_{in}} = \frac{200 \times 5 \times 10^3}{1000} = 1000$$



Q26. Define the term 'critical frequency' in relation to sky wave propagation of electromagnetic waves. On a particular day, the maximum frequency reflected from the ionosphere is 10 MHz. On another day, it was found to decrease to 8 MHz. Calculate the ratio of the maximum electron densities of the ionosphere on the two days.

Answer: Definition: Critical frequency is the **highest frequency** of the radio wave which when sent normally towards the given layer of the ionosphere gets reflected from ionosphere and return to the earth.

Critical frequency is given by, $f_c = a (N_{\max})^{1/2}$

Where $N_{\max}$ is the maximum electron density of the ionosphere layer.

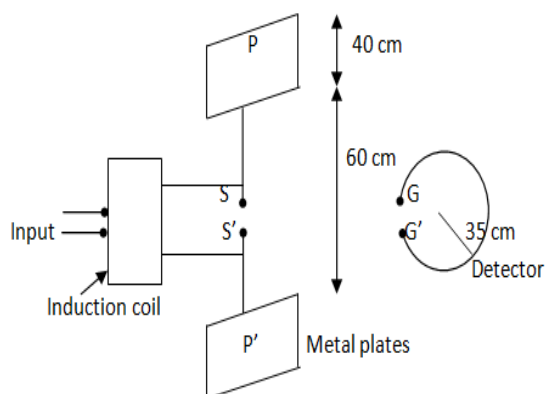
Problem:

$$f_c = 10 \text{ MHz} = a(N_{\max})^{1/2} \quad \text{and} \quad f'_c = 8 \text{ MHz} = a(N'_{\max})^{1/2}$$

$$\frac{N_{\max}}{N'_{\max}} = \left(\frac{f_c}{f'_c}\right)^2 = \left(\frac{10}{8}\right)^2 = 25:16$$

Q27. Draw a labeled diagram of Hertz's experimental set-up to produce electromagnetic waves. Explain the generation of electromagnetic waves using this set-up.

Answer:

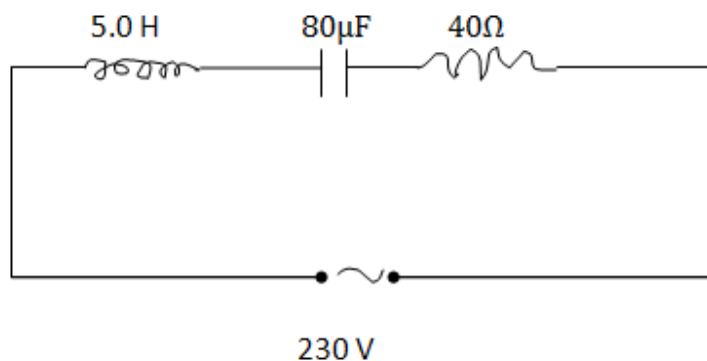


Experimental setup: Two large metal plates P and P' are connected to metal spheres S and S' as shown. The spheres are connected to an induction coil. High voltage is set-up across the gap by interrupting current in the coils. This voltage ionizes the air in the gap which produces oscillating current in the gap SS'.

This produces electromagnetic waves. For detection of these electromagnetic waves there is a detector which consists of a single loop of wire connected to spheres G and G'.



Q28. The given circuit diagram shows a series LCR circuit connected to a variable frequency 230 V source:



- Determine the source frequency which drives the circuit in resonance.
- Obtain the impedance of the circuit and the amplitude of current at the resonating frequency.
- Determine the rms potential drops across the three elements of the circuit.
- How do you explain the observation that the algebraic sum of the voltages across the three elements obtained in (c) is greater than the supplied voltage?

Answer:

(a) We know that resonance frequency is given by $\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50 \times 80 \times 10^{-6}}} = 50 \text{ rad s}^{-1}$

b) At resonance, $Z = R = 40 \Omega$

Current amplitude,

$$I_p = \frac{E_p}{Z} = \frac{E_p}{R} = \frac{\sqrt{2}E_{rms}}{R} = \frac{1.414 \times 230}{40} = 8.1 \text{ A}$$

c) Current through the circuit $I_{rms} = \frac{E_{rms}}{R} = \frac{230}{40} = \frac{23}{4} \text{ A}$

Potential drop across the inductance:

$$V_{Lrms} = I_{rms} \times \omega_r L = \frac{23}{4} \times 50 \times 5 = 1437.5 \text{ V}$$

Potential drop across the capacitor:

$$V_{Crms} = I_{rms} \times \frac{1}{\omega_r C} = \frac{23}{4} \times \frac{1}{50 \times 80 \times 10^{-6}} = 1437.5 \text{ V}$$

Potential drop across the resistance:

$$V_{Rrms} = I_{rms} R = \frac{23}{4} \times 40 = 230 \text{ V.}$$

- (d) Thus we find that the algebraic sum of the three voltages is more than the source voltage of 230V. These voltages are not in the same phase for example current leads potential difference in capacitor where as current lags potential in inductance therefore potential cannot be added like ordinary numbers. The voltages across L and C are out of phase hence results to zero. So $V_{Rrms} = \text{Applied rms voltage.}$



Or,

The primary coil of an ideal step-up transformer has 100 turns and the transformation ratio is also 100. The input voltage and the power are 220V and 1100W respectively. Calculate

- (I) Number of turns in the secondary
- (II) The current in the primary
- (III) Voltage across the secondary
- (IV) Power in the secondary
- (V) The current in the secondary

Answer: Given Number of turns in Primary = $N_p = 100$, Transformation ratio = 100,

Voltage across Primary = $V_p = 220$ V

Power across Primary = $P_p = 1100$ W

Also Transformation Ratio = No of turns in secondary / No of turns in primary.

This ratio then directly corresponds to the ratio of secondary voltage (output) to primary voltage (input).

- (I) $N_s = \text{Transformation ratio} \times N_p = 100 \times 100 = 10000$.
- (II) $I_p = \frac{P_p}{V_p} = \frac{1100}{220} = 5$ A
- (III) $V_s = \text{Transformation ratio} \times V_p = 100 \times 220 = 22000$ V
- (IV) $P_s = P_p = 1100$ W
- (V) We know $P = I \times V$ therefore $I_s = \frac{P_s}{V_s} = \frac{1100}{22000} = .05$ A

Q29. What is the interference of light? Write two essential conditions for sustained interference pattern to be produced on the screen.

Draw a graph showing the variation of intensity versus the position on the screen in Young's experiment when (a) both the slits are opened and (b) one of the slit is closed.

What is the effect on the interference pattern in Young's double slit experiment when:

- (I) Screen is moved closer to the plane of slits?
- (II) Separation between two slits is increased.

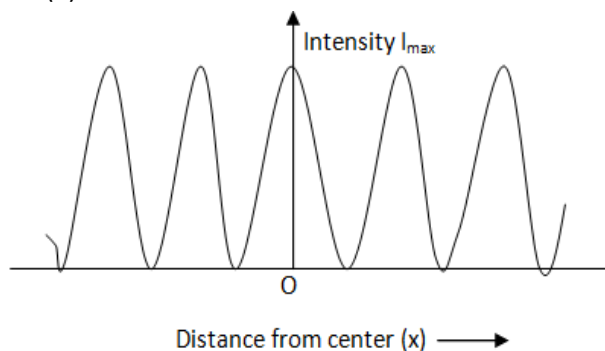
Explain your answer in each case.

Answer: The modification in the intensity distribution due to the superposition of two or more coherent wave is known as **Interference**. On the screen we get alternative bright and dark region known as interference of fringes.

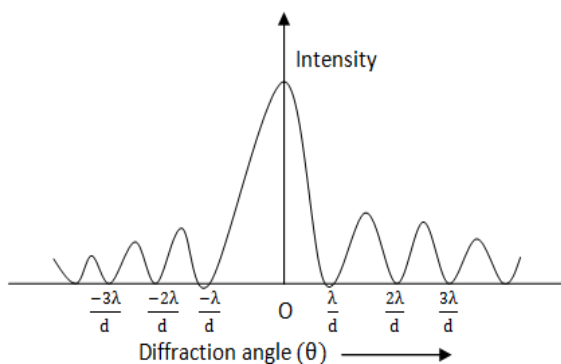


Conditions for sustained interference:

- (I) The two sources of light must be coherent.
- (II) The two sources must be narrow



Position is taken along X-axis and Intensity is along Y-axis when both the slits are open, following interference pattern observed on the screen.



When one of the slits is closed, diffraction pattern is obtained on the screen. The following intensity curve is obtained.

Fringe width, $\beta = \frac{D\lambda}{d}$ from this formula it is evident that fringe width proportional to D and inversely proportional to d .

- (I) When screen is moved closer to the plane of the slits, the distance D decreases, the fringe width β also decreases.
- (II) When separation d between two slits is increased, fringe width β decreases.



Or,

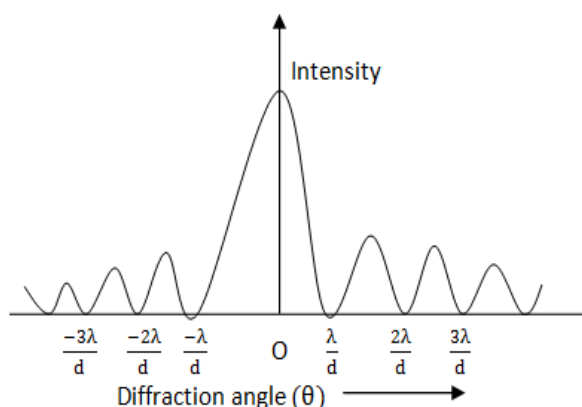
What is diffraction of light? Draw a graph showing the variation of intensity with angle in a single slit diffraction experiment. Write one feature which distinguishes the observed pattern from the double slit interference pattern.

How would the diffraction pattern of a single slit be affected when:

- (I) The width of the slit is decreased.
- (II) The monochromatic source of light is replaced by a source of white light.

Answer: The bending of light round a sharp edge or a narrow opening or a very small obstacle is known as **diffraction**. By sharp narrow or small we mean that these sizes should be comparable to the wave length of light.

The intensity distribution wave for diffraction:



In interference, all bright fringes are of same intensity however in diffraction, the intensity of bright fringes decreases with the increase in distance from the central bright fringe.

- (I) When the width of the slit is decreased, the diffraction pattern becomes narrower.
- (II) When monochromatic source is replaced by white light source, we get a colored diffraction pattern with central white band also fringe widths are different for example red fringe is wider than the violet fringe.

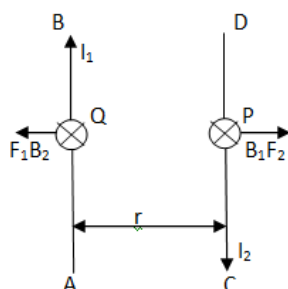


Q30. Derive a mathematical expression for the force per unit length experienced by each of the two long current carrying conductors placed parallel to each other in air. Hence define one ampere of current. Explain why two parallel straight conductors carrying current in the opposite direction kept near each other in air repel?

Answer: Considering two infinitely long thin conductors carrying currents in opposite directions.

B_1 = Magnetic field produced at P on the conductor CD due to I_1 current at conductor AB

$$B_1 = \frac{\mu_0 I_1}{2\pi r}$$



The magnetic field B_1 is perpendicular to plane of paper and directed inward.

F_2 = Force per length on CD due to B_1

$$F_2 = B_1 I_2 = \frac{\mu_0 I_1 I_2}{2\pi r} \quad [\because F = BIl, \text{ Since unit length } l = 1]$$

By *Fleming's left hand rule* direction of F_2 is away from the conductor AB.

Similarly the current I_2 will create a field B_2 at Q directed inward which in turn will create force/length F_1

$$\therefore F_1 = B_2 I_1 = \frac{\mu_0 I_1 I_2}{2\pi r}$$

By *Fleming's left hand rule*, the direction of F_1 is away from the conductor AB.

Thus the direction F_1 and F_2 on the conductors opposite hence repel each other.

Definition of Ampere: From the above expression $F = \frac{\mu_0 I_1 I_2}{2\pi r}$

$$\text{If } I_1 = I_2 = 1 \text{ A, and } r = 1 \text{ m, then } F = \frac{\mu_0}{2\pi} = \frac{4\pi \times 10^{-7}}{2\pi} = 2 \times 10^{-7} \text{ Nm}^{-1}$$

Thus **one ampere** is that current which on flowing through each of the two parallel uniform linear conductors placed in free space at a distance of one meter from each other produces between them a force of $2 \times 10^{-7} \text{ N}$ per meter of their lengths.